

Student’s t-distribution:

The static ‘ t ’ was introduced by W S.Gosset in 1908 who wrote under the name “Student”. That is why it is called student's t test. Later on its distribution was rigorously established by Prof. R.A. Fisher in 1926. It can used when the population standard deviation is not known and the size of the sample is less than or equal to thirty.

A random variable X is said to follow t-distribution, if its probability density function is given by

$$f(t) = \frac{k}{\left(1 + \frac{t^2}{v}\right)^{v+1/2}}, \quad -\infty < t < \infty$$

where v is known as the degrees of freedom and k is constant. The constant value k is chosen in

such a way that $\int_{-\infty}^{\infty} f(t)dt = 1$. After simplification, we get $k = \frac{1}{\sqrt{v}\beta\left(\frac{1}{2}, \frac{v}{2}\right)}$.

Assumption of t-distribution:

- 1) The population from which the sample is drawn is normal.
- 2) The sample is random and size $n \leq 30$.
- 3) The population S.D. σ is not known.

Properties of t-distribution:

- 1) The probability curve of t-distribution is symmetrical.
- 2) The tails of the curve are asymptotic to x-axis.
- 3) When $n \rightarrow \infty$, t- distribution tends to normal distribution.
- 4) The form of the t-dist. Varies with the degrees of freedom.

Application of t- distribution: The t- distribution is used

- 1) To test significance of the mean of sample.
- 2) To test the difference between two means or to compare two samples.

***t*-Test for Single Mean.** Suppose we want to test :

(i) if a random sample x_i ($i = 1, 2, \dots, n$) of size n has been drawn from a normal population with a specified mean, say μ_0 , or

(ii) if the sample mean differs significantly from the hypothetical value μ_0 of the population mean.

Under the null hypothesis H_0 :

(i) The sample has been drawn from the population with mean μ or (ii) there is no significant difference between the sample mean $\bar{x}$ and the population mean μ .

the statistic

$$t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$,

follows Student's t -distribution with $(n-1)$ d.f.

We now compare the calculated value of t with the tabulated value at certain level of significance. If calculated $|t| > \text{tabulated } t$, null hypothesis is rejected and if calculated $|t| < \text{tabulated } t$, H_0 may be accepted at the level of significance adopted.

Relationship between the sample variance (s) and S :-

Let (s) be the sample variance then

$$\frac{S^2}{n} = \frac{s^2}{(n-1)}$$

$$z = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}} \right)} = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n-1}} \right)}$$

The mean lifetime of a sample of 25 bulbs is found as 1550 hours with a SD of 120 h. The company manufacturing the bulbs claims that the average life of their bulbs is 1600 h. Is the claim acceptable at 5% LOS?

Soln:-

Null Hypothesis (H_0): - The average life time of the bulb manufactured by the company and the sample is not significant
($\bar{x} = \mu$)

Alternative Hypothesis H_1 : - $\bar{x} < \mu$.

This Hypothesis represents a left tailed test.

Test statistic
$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}}$$

Here $\bar{x} = 1550$; $\mu = 1600$; $s = 120$; $n = 25$.

$$t = \frac{1550 - 1600}{\frac{120}{\sqrt{24}}} = -2.04.$$

degree of freedom $\gamma = n - 1 = 25 - 1 = 24$.

From the t-test table $t_{0.05} = 1.71$

Since $|t| > |t_{0.05}|$ we reject the null hypothesis.

Ten individuals are chosen at random from a population and their heights are found to be in inches 63, 63, 64, 65, 66, 69, 69, 70, 70, 71. Discuss the suggestion that the mean height in the universe is 65 inches given that for 9 degrees of freedom the value of Student's t and 5 percent level of significance is 2.262.

Soln:- We first find the sample mean and sample standard deviation.

x	$(x - \bar{x})$ $x - \bar{x}$	$(x - \bar{x})^2$ $(x - \bar{x})^2$
63	-4	16
63	-4	16
64	-3	9
65	-2	4
66	-1	1
69	2	4
69	2	4
70	3	9
70	3	9
71	4	16
$\sum x = 670$		$\sum (x - \bar{x})^2 = 88$

Hence $\bar{x} = 670/10 = 67$.

Sample standard deviation (s) = $\sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} = \sqrt{\frac{88}{9}} = 3.13$

Null Hypothesis: The mean height of the universe is 65

Alternative Hypothesis: The mean height is greater than 65 ($\mu > 65$)

This hypothesis represents a right tailed test

test statistic $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}} = \frac{67 - 65}{\frac{3.13}{\sqrt{9}}} = \frac{6}{3.13} = 1.92$

Given $t_{0.05} = 2.262$

Since $|t| < t_{0.05}$ we accept the null hypothesis.

t-Test for Difference of Means. Suppose we want to test if two independent samples x_i ($i = 1, 2, \dots, n_1$) and y_j ($j = 1, 2, \dots, n_2$) of sizes n_1 and n_2 have been drawn from two normal populations with means μ_X and μ_Y respectively.

Under the null hypothesis (H_0) that the samples have been drawn from the normal populations with means μ_X and μ_Y and under the assumption that the population variance are equal, i.e., $\sigma_X^2 = \sigma_Y^2 = \sigma^2$ (say), the statistic

$$t = \frac{(\bar{x} - \bar{y}) - (\mu_X - \mu_Y)}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where

$$\bar{x} = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i, \quad \bar{y} = \frac{1}{n_2} \sum_{j=1}^{n_2} y_j$$

and

$$S^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum_i (x_i - \bar{x})^2 + \sum_j (y_j - \bar{y})^2 \right]$$

is an unbiased estimate of the common population variance σ^2 , follows Student's t -distribution with $(n_1 + n_2 - 2)$ d.f.

Two independent samples of 8 and 7 items respectively had the following values of the variable (weight in ounces):

Sample 1: 9 11 13 11 15 9 12 14

Sample 2: 10 12 10 14 9 8 10

Is the difference between the means of the sample significant? Given $t_{0.05} = 2.16$.

Soln:-

Null Hypothesis: (H_0) : The mean difference between the given two samples is not significant.

Alternative Hypothesis (H_1) : The mean difference between the given two samples is significant.

(Assumed mean)

x	$x - 11.75$	$(x - 11.75)^2$	y	$(y - 10.43)$	$(y - 10.43)^2$
9	-2.75	7.5625	10	-0.43	0.1849
11	-0.75	0.5625	12	1.57	2.4649
13	1.25	1.5625	10	-0.43	0.1849
11	-0.75	0.5625	14	3.57	12.7449
15	3.25	10.5625	9	-1.43	2.0449
9	-2.75	7.5625	8	-2.43	5.9049
12	0.25	0.0625	10	-0.43	0.1849
14	2.25	5.0625			

$$\sum x = 94 \quad \sum (x - 11.75)^2 = 33.5 \quad \sum y = 73 \quad \sum (y - 10.43)^2 = 23.7143$$

$$\therefore \bar{x} = \frac{94}{8} = 11.75 \quad ; \quad \bar{y} = \frac{73}{7} = 10.43$$

$$S^2 = \frac{1}{8+7-2} \left[\sum (x_i - \bar{x})^2 + \sum (y_j - \bar{y})^2 \right] = \frac{1}{13} [33.5 + 23.7143]$$

$$S^2 = 4.4011 \quad \therefore S = 2.09787$$

$$\text{test statistic } t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{11.75 - 10.43}{2.09787 \sqrt{\frac{1}{8} + \frac{1}{7}}}$$

$$= 1.2157$$

$$\text{Given } t_{0.05} = 2.16$$

Since $|t| < |t_{0.05}|$ we accept the null hypothesis.